

HONGBO WEN

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RESEARCH INTERESTS

I develop program analysis, synthesis, and verification techniques to ensure the soundness and correctness of computation across diverse algebraic systems, including finite-field constraints in zero-knowledge proof systems (ZKP), nonlinear arithmetic in financial protocols, and cryptographic protocols.

I am also interested in the full lifecycle of LLM systems for coding tasks, viewing reliable coding capabilities as a foundation for increasingly general-purpose AI systems.

Soundness and Correctness across Diverse Algebraic Systems. Many systems rely on algebraic representations of computation to provide verifiability, privacy, scalability, or asynchrony. Bugs are difficult to find because computation moves across several representations that can each look correct in isolation, yet their mismatch can accept invalid results, reject valid ones, or compromise funds. I develop program analysis for ZK circuits and zkVMs [USENIX Security'24, NDSS'27], program synthesis for efficient ZK circuits and financial-protocol attacks [OOPSLA'25, CCS'24], and verification-guided design for cryptographic payment protocols [CCS'26]. I am extending these ideas to verify asynchronous floating-point computation in distributed LLM training.

LLM Systems for Coding Tasks. Reliable coding is a key step toward increasingly general-purpose AI because it combines open-ended reasoning, real-world agency, and executable feedback. I study application-layer security for coding agents and program-synthesis techniques for generating task-specific coding data and agent structures.

EDUCATION

University of California, Santa Barbara, CA, USA

Sep 2022 - Present

Ph.D. candidate in Computer Science

Advisor: Yu Feng

GPA 4.0/4.0

Tsinghua University, Beijing, China

Aug 2015 - Jun 2020

Bachelor of Engineering

PUBLICATIONS

- [1] **Hongbo Wen**, Jingyu Ke, Hanzhi Liu, Yanju Chen, Yu Feng. *Revealing Semantic Gaps in Zero-Knowledge Virtual Machines via Obligation-Targeted Semantic Fuzzing*. **NDSS 2027**.
- [2] **Hongbo Wen**, Hanzhi Liu, Yanju Chen, Jingyu Ke, Dahlia Malkhi, Yu Feng. *Thunderbolt: Fast Asynchronous Off-Chain Bitcoin Transfers*. **ACM CCS 2026**.
- [3] Junrui Liu, Jiaxin Song, Yanning Chen, Hanzhi Liu, **Hongbo Wen**, Luke Pearson, Yanju Chen, Yu Feng. *Tabby: A Synthesis-Aided Compiler for High-Performance Zero-Knowledge Proof Circuits*. **OOPSLA 2025**.
- [4] **Hongbo Wen**, Hanzhi Liu, Jiaxin Song, Yanju Chen, Wenbo Guo, Yu Feng. *FORAY: Towards Effective Attack Synthesis against Deep Logical Vulnerabilities in DeFi Protocols*. **ACM CCS 2024**.
- [5] **Hongbo Wen**, Jon Stephens, Yanju Chen, Kostas Ferles, Shankara Pailoor, Kyle Charbonnet, Isil Dillig, Yu Feng. *Practical Security Analysis of Zero-Knowledge Proof Circuits*. **USENIX Security 2024**.

INVITED TALKS

Tabby: A Synthesis-Aided Compiler for High-Performance Zero-Knowledge Proof Circuits

SoCalPLS, USC, Los Angeles, CA, USA

Feb 2026

Tabby: Automated Programming of Efficient Zero-Knowledge Proof Circuits

ETHDenver, Denver, CO, USA

Feb 2025

Practical Security Analysis of Zero-Knowledge Proof Circuits

SoCalPLS, UCSD, San Diego, CA, USA

Feb 2025

FORAY: Towards Effective Attack Synthesis against Deep Logical Vulnerabilities in DeFi Protocols

ACM CCS '24, Salt Lake City, UT, USA

Oct 2024

Practical Security Analysis of Zero-Knowledge Proof Circuits

USENIX Security '24, Philadelphia, PA, USA

Aug 2024

GRANTS AND AWARDS

Ethereum Foundation Academic Grants Round 2024 Yu Feng, Hanzhi Liu, Hongbo Wen <i>Topic: Sentinel: Adaptive Counter-Attack Synthesis for Mitigating Onchain Exploits</i>	Jun 2024
Ethereum Foundation Academic Grants Round 2023 Yu Feng, Yanju Chen, Hongbo Wen <i>Topic: Financial Model-Driven Attack Synthesis for DeFi</i>	Jun 2023
Academic Excellence Fellowship, UC Santa Barbara	2022

RESEARCH-INDUSTRY INTEGRATION

Riema Labs Inc. Co-founder & Principal Architect. • Led applied AI research and its adoption in development workflows across a 50-person engineering organization, mentoring teams and shipping security-critical systems for 150K+ users. • Advanced <i>Thunderbolt</i> research alongside Bitcoin-native product development, translating cryptographic protocol design into deployable infrastructure [CCS'26].	Jan 2024 - Present
Veridise Inc. Research & Development Engineer, Smart Contract and Zero-Knowledge Security. • Developed <i>ZKAP</i> and <i>FORAY</i> from first-hand security insights and data gathered through real-world industry engagements, turning recurring industry problems into research questions [USENIX Security'24, CCS'24]. • Applied the resulting program-analysis and synthesis techniques back to practice, discovering 30+ critical vulnerabilities in widely used ZK projects and 10+ zero-day vulnerabilities in DeFi protocols.	Mar 2022 - Aug 2022

SERVICE

USENIX Security '25 Artifact Evaluation Committee	2025
CS162 Programming Languages , Teaching Assistant, UC Santa Barbara	2023 – 2025